



MIKROSKIL IDEAFUSE COMPETITION 2025

X-File

You are a royal surveyor mapping ancient battlefields. Each battlefield forms a strange, **12-pointed "X"-shaped polygon**, drawn carefully on graph paper where all coordinates are **integers**.

The king wants a detailed map showing how many **integer lattice points** are *inside* each X-shaped boundary and how many lie *on* it.

For each battlefield (polygon), determine:

1. The number of **lattice points strictly inside** the polygon.
2. The number of **lattice points on its boundary**.

Input Format

- The first line contains an integer T — number of polygons ($1 \leq T \leq 10^4$)
- For each test case has **exactly 12 vertices**, given in order (either clockwise or counterclockwise).
- Each vertex (x_i, y_i) has integer coordinates satisfying $|x_i|, |y_i| \leq 10^9$.

Output Format

- For each test case, print one result per line.
- $I \ B$ — I number of **lattice points strictly inside** the polygon. B — number of **lattice points on the boundary**.

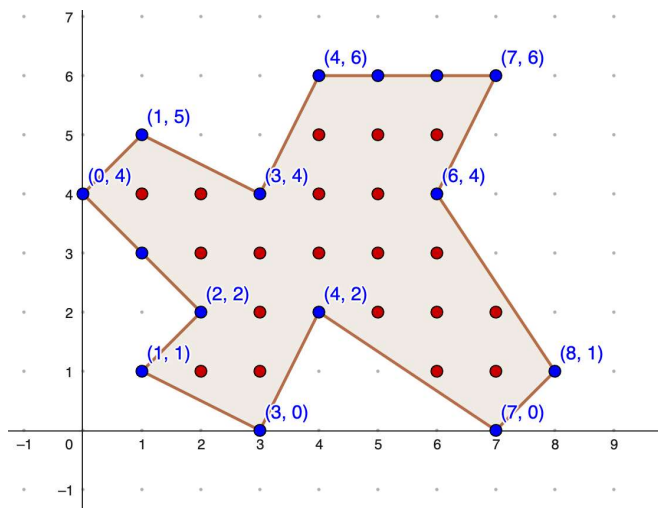
Sample Input 1

```
1
1 1
3 0
4 2
7 0
8 1
6 4
7 6
4 6
3 4
1 5
0 4
2 2
```

Sample Output 1

Case #1: 20 15

Explanation



- Red points: points inside polygon
- Blue points: points on the boundary